Exercise 02. SQL FOUNDATIONS

01 TABLE students

student_id	name	age	department
1	Alice	20	IT
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

01.
```
SELECT DISTINCT (department)
       FROM students;
```

department
IT
HR
Finance

02
```
SELECT department,
       AVG(age) AS avg_age
       FROM students
       GROUP BY department;
```

department	avg_age
IT	20.5
HR	22
Finance	23

03
```
SELECT department,
       COUNT(student_ID) AS student_count
       FROM students
       GROUP BY department
       HAVING student_count > 1;
```

department	student_count
IT	2
HR	2

04
```
SELECT student_id,
       name,
       age,
       department
FROM students
WHERE age BETWEEN 21 AND 23;
```

student_id	name	age	department
3	Charlie	21	IT
2	Bob	22	HR
5	Eve	22	HR
4	Diana	23	Finance

05
```
SELECT student_id,
       name,
       age,
       department
FROM students
WHERE (department = 'IT' OR department = 'HR'
AND age > 21;
```

student_id	name	age	department
2	Bob	22	HR
3	Eve	22	HR

02 The courses Table

courses

course_id	course_name	department	credits
101	SQL Basics	IT	3
102	Python	IT	4
103	Data Science	IT	4
104	Excel	Finance	2
105	Statistics	HR	3

06
```
SELECT department,
       sum(credits) AS total_credits
FROM courses
GROUP BY department
HAVING total_credits > 5;
```

department	total_credits
IT	11

07
```
SELECT course_id,
       course_name,
       department,
       credits
FROM courses
WHERE NOT credits = 4;
```

course_id	course_name	department	credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

08
```
SELECT course_id,
       course_name,
       credits
FROM courses
ORDER BY credits DESC
LIMIT 3;
```

course_id	course_name	credits
102	Python	4
103	Data Science	4
101	SQL Basics	3

03 The enrollments table

enrollments

enrollment_id	student_id	course_id	grade
1	1	101	85
2	2	102	78
3	3	103	90
4	4	104	88
5	5	105	82

09
```
SELECT MAX(grade) AS max_grade,
       MIN(grade) AS min_grade,
       AVG(grade) AS avg_grde
  FROM enrollments;
```

max_grade	min_grade	avg_grade
90	78	84.6

10
```
SELECT course_id;
       COUNT(enrollment_id) AS enrollment count
  FROM enrollments
  GROUP BY course_id;
```

course_id	enrollment_count
101	1
102	1
103	1
104	1
105	1

04 The salaries Table

Salaries

employee_id	name	department	salary	bonus
1	Tom	IT	60000	5000
2	Jerry	HR	55000	4000
3	Spike	Finance	70000	6000
4	Tyke	IT	62000	5500
5	Butch	HR	54000	3500

11.
```
SELECT department,
       sum(salary) AS total_salary,
       sum(bonus) AS total_bonus,
FROM salaries
GROUP BY department;
```

department	total_salary	total_bonus
IT	122 000	10 500
HR	109 000	7500
Finance	70000	6000

12.
```
SELECT department,
       AVG(salary) AS avg-salary
FROM salaries
GROUP By department
HAVING avg_salary > 55 000;
```

department	avg-salary
IT	61 000
Finance	70000

13.
```
SELECT employee_id,
       name,
       salary,
       bonus
   salary + bonus AS total_compensation
 FROM salaries
WHERE total_compensation > 60 000;
```

employee_id	name	salary	bonus	total_compensation
1	Tom	60000	5000	65000
3	Spike	70000	6000	76000
4	Tyke	62 000	5500	67500
5				

05 The projects table

projects

project_id	project_name	department	budget
1	AI APP	IT	120 000
2	Payroll System	Finance	80 000
3	Dashboard	IT	150 000
4	Website	Marketing	60 000
5	HR Portal	HR	50 000

14.
```
SELECT department,
       sum(budget) AS total_budget,
       AVG(budget) AS avg-budget
FROM projects
GROUP BY department
HAVING avg-budget > 70 000;
```

department	total_budget	avg-budget
IT	270000	135 000
Finance	80 000	80 000

15. SELECT project_id,
project_name,
department,
budget
FROM projects
WHERE budget BETWEEN 50 000 AND 120 000
AND department NOT ('Marketing');

project_id	project_name	department	budget
1	AI APP	IT	120000
2	Payroll system	Finance	80 000
5	HR Portal	HR	50 000